

Online Retail Transactions

Customer Purchase Behaviour Dashboard

2010–2011 | Power BI Desktop

£8.91M Revenue · 5.17M Units · 397,884 Clean Records · 38 Countries · 4,338 Customers

TOOL	SOURCE	RECORDS	DATE RANGE
Power BI Desktop	UCI Repository	397,884 · 9 Cols	Dec 2010–Dec 2011

01 EXECUTIVE SUMMARY

- Processed 541,909 raw retail transaction records through a 5-step Power Query ETL pipeline — removing 9,288 cancelled orders (C-prefix invoices), 135,080 null-customer rows, 10,624 negative quantity rows, 2,517 zero-price rows, and 1,454 null description rows — retaining 397,884 clean records (73.4% retention rate) across 38 countries.
- Created a Revenue column (Quantity × UnitPrice) in Power Query, corrected 2 data type errors (StockCode to Text, InvoiceDate to Date), engineered 5 DAX measures (Total Revenue, Total Quantity, Unique Customers, Unique Products, Avg Order Value), and built a 12-visual interactive dashboard with 2 cross-filtering slicers.
- Analyzed £8,911,407.90 total revenue across 5,167,812 units, 4,338 unique customers, 18,532 invoices, and 3,665 products — identifying United Kingdom as the dominant market at £7,308,391 (82.01% of total revenue) across 3,920 customers and 16,646 orders.
- Identified November 2011 as peak revenue month at £1,161,817 (Christmas shopping surge — 159.7% above February 2011 trough of £447,137), Paper Craft Little Birdie as top product at £168,469 (1.89% share), and Avg Order Value of £480.87 across all 18,532 invoices.

02 PROBLEM STATEMENT

An online UK-based retailer's transactional data for 2010–2011 existed as a flat Excel file with 541,909 raw records containing significant data quality issues — cancelled orders mixed with genuine sales, transactions with no customer attribution, negative quantities from returns, zero-price test entries, and no pre-calculated revenue column. With no dashboard or reporting layer, management had no visibility into which products drove revenue, which countries represented growth opportunities, how purchasing volumes trended month-to-month, or what the peak seasonal demand windows were — leaving a £8.91M revenue portfolio entirely unanalyzed.

03 OBJECTIVES

- Obj 1: Execute a 5-step ETL pipeline in Power Query — removing cancelled orders, null customers, invalid quantities and prices, and null descriptions from 541,909 raw records.
- Obj 2: Engineer a Revenue column (Quantity × UnitPrice) as a calculated column in Power Query with error protection.
- Obj 3: Create 5 DAX measures enabling dynamic aggregation of revenue, quantity, customer counts, product counts, and average order values.

- Obj 4: Build a 12-visual dashboard covering KPIs, country comparison, product rankings, monthly trend, geographic map, and product table.
- Obj 5: Enable cross-filtering via 2 slicers (Country, InvoiceDate range) for multi-dimensional analysis.
- Obj 6: Surface actionable insights — UK market concentration risk, seasonal demand peaks, top revenue-generating products, and international growth opportunities.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Online Retail Dataset
Source	UCI Machine Learning Repository — archive.ics.uci.edu/dataset/352/online+retail
File Format	Microsoft Excel Workbook (.xlsx)
Raw Records	541,909 rows — before cleaning
Clean Records	397,884 rows — after removing invalid/cancelled/null data (73.4% retention)
Date Coverage	December 1, 2010 – December 9, 2011 (13 months)
Geographic Scope	38 countries worldwide
Data Issues	Cancelled orders, null CustomerIDs, negative quantities, zero prices, null descriptions
Key Derived Column	Revenue = Quantity × UnitPrice (created in Power Query Add Column)

Column Definitions (9 Columns after ETL)

Column	Type	Description
InvoiceNo	Text	Unique invoice number — cancelled invoices begin with 'C' (removed in ETL)
StockCode	Text	Product code — mix of numeric and alphanumeric values
Description	Text	Product name — 3,665 unique products after cleaning
Quantity	Whole Number	Units per transaction — negatives removed (returns/errors)
InvoiceDate	Date	Transaction date — Dec 2010 to Dec 2011 (changed from Date/Time)
UnitPrice	Decimal	Price per unit in GBP — zero-price rows removed
CustomerID	Text	Unique customer identifier — 4,338 unique after removing nulls
Country	Text	Customer's country — 38 unique countries, UK dominates at 82.01%
Revenue	Decimal	Calculated: Quantity × UnitPrice — created in Power Query

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Power BI Desktop	Microsoft (Latest 2024)	Primary BI tool — ETL, calculated column, DAX, dashboard
Power Query Editor	Built into Power BI	5-step ETL: remove cancelled, nulls, negatives, zeros, add Revenue column
DAX	Power BI Native	5 measures: Revenue, Quantity, Customers, Products, Avg Order Value
Excel (.xlsx)	Source Format	Raw dataset from UCI — imported via Get Data → Excel Workbook
UCI Repository	Academic Dataset Platform	Public dataset: archive.ics.uci.edu/dataset/352/online+retail
JSON Theme File	Power BI Custom Theme	Digital Commerce — teal/navy/white e-commerce branded theme

06 ETL PROCESS — POWER QUERY (8 STEPS)

The raw dataset contained 541,909 rows with 4 major data quality issues that would corrupt any analysis if not removed. All 8 ETL steps were executed in Power Query Editor before data entered the model, creating a clean 397,884-row dataset with a 73.4% retention rate. Every step is logged in the Applied Steps pane for full auditability.

Step	Power Query Action	Rows Affected / Reason
01	Get Data → Excel Workbook → selected 'Online Retail' sheet	Loaded 541,909 raw rows into Power Query Editor
02	Clicked 'Transform Data' to open Power Query Editor	Enabled pre-load cleaning — never let dirty data enter the model
03	InvoiceNo → Text Filters → Does Not Begin With → 'C'	Removed 9,288 cancelled orders — C-prefix = cancellation, not a sale
04	CustomerID → uncheck 'null' in dropdown filter	Removed 135,080 anonymous transactions with no customer attribution
05	Quantity → Number Filters → Greater Than → 0	Removed 10,624 return/adjustment rows with negative quantities
06	UnitPrice → Number Filters → Greater Than → 0	Removed 2,517 test/error rows with zero prices
07	StockCode type → Text; InvoiceDate type → Date	Fixed mixed alignment (StockCode) and removed time from dates
08	Add Column → Custom Column → Revenue = [Quantity]*[UnitPrice]	Created Revenue column; changed type to Decimal Number
09	Home → Close & Apply	Loaded 397,884 clean rows — 0 errors, all types correct

07 DAX MEASURES CREATED

Measure	DAX Formula	Result / Use
Total Revenue	SUM('Online Retail'[Revenue])	£8,911,407.90 — primary revenue KPI
Total Quantity	SUM('Online Retail'[Quantity])	5,167,812 — volume KPI card
Unique Customers	DISTINCTCOUNT('Online Retail'[CustomerID])	4,338 — customer reach metric
Unique Products	DISTINCTCOUNT('Online Retail'[StockCode])	3,665 — catalogue size metric
Avg Order Value	DIVIDE([Total Revenue],DISTINCTCOUNT('Online Retail'[InvoiceNo]))	£480.87 — avg basket size

08 DASHBOARD VISUALS BUILT (12 Visuals)

#	Visual Type	Fields Used	Insight Delivered
1	KPI Card	Total Revenue	£8,911,407.90 total retail revenue
2	KPI Card	Total Quantity	5,167,812 total units sold
3	KPI Card	Unique Customers	4,338 unique customers
4	KPI Card	Unique Products	3,665 unique products
5	Clustered Bar Chart	Y: Country / X: Total Revenue	UK dominates at £7.31M (82.01%)
6	Line Chart	X: InvoiceDate / Y: Total Revenue	Nov 2011 peak £1.16M — Christmas surge
7	Donut Chart	Legend: Country / Values: Revenue	UK 82% slice vs all other countries 18%
8	Map Visual	Location: Country / Color: Revenue	Global revenue heat map — UK dark, others pale
9	Table	Description, Total Revenue, Total Quantity	Top 10 products with revenue and volume
10	Slicer	Country (38 values)	Filters all visuals by country
11	Slicer	InvoiceDate (date range slider)	Filters all visuals by date range
12	Bar Chart	Y: Description / X: Total Revenue	Top products ranked by revenue

09 KEY INSIGHTS & RESULTS

A. Revenue & Volume Overview

- Analyzed £8,911,407.90 total revenue across 397,884 clean transactions, 5,167,812 units sold, 18,532 invoices, 4,338 unique customers, and 3,665 unique products — averaging £480.87 per order and £3.12 per unit across 13 months of trading.

- Identified 2011 as the primary trading year at £8,338,694 (93.6% of total revenue) vs December 2010 at £572,714 (6.4%) — confirming the dataset is predominantly a full-year 2011 analysis with one month of 2010 baseline data.

B. Country Analysis — Heavy UK Concentration

- United Kingdom generated £7,308,391 (82.01% of total revenue) across 3,920 customers and 16,646 orders — creating a single-market dependency risk where any UK demand disruption would eliminate over 4/5 of total business revenue.
- Netherlands ranked #2 at £285,446 (3.20%) with only 9 customers — indicating a high-value wholesale relationship; EIRE #3 at £265,546 (2.98%) with only 3 customers generating 260 orders — flagging potential high-frequency institutional buyers outside the UK.
- Germany (#4, £228,867 — 2.57%) and France (#5, £209,024 — 2.35%) represent the largest retail growth opportunities with 94 and 87 customers respectively — the most diversified non-UK markets with genuine consumer bases.

C. Monthly Revenue Trend & Seasonality

- November 2011 peaked at £1,161,817 (2,657 orders) — the clear Christmas shopping surge; October 2011 at £1,039,319 (1,929 orders) ranked second — confirming a Q4 demand acceleration pattern beginning in September 2011 at £952,838.
- February 2011 registered as the trough at £447,137 (997 orders) — 159.7% below the November peak, representing the classic post-Christmas retail demand collapse; YoY monthly comparison impossible as Dec 2010 (£572,714) is the only 2010 data point.
- September–November 2011 combined generated £3,153,973 — 35.4% of full-year revenue in just 3 months — confirming extreme Q4 seasonal concentration that requires inventory and staffing preparation from August onwards.

D. Product Performance

- Paper Craft Little Birdie ranked #1 at £168,469 (1.89% share, 80,995 units) from a single order — representing a bulk wholesale purchase; Regency Cakestand 3 Tier ranked #2 at £142,593 across 1,703 individual orders — the most consistently demanded retail product.
- White Hanging Heart T-Light Holder ranked #3 at £100,448 across 1,971 orders — highest order frequency of any product, appearing in over 10.6% of all 18,532 invoices; Jumbo Bag Red Retrospot at £85,221 (46,181 units, 1,600 orders) showed strong repeat-purchase behaviour.
- Top 10 products collectively generated £926,024 — 10.4% of total revenue — confirming a highly fragmented catalogue (3,665 products) with no single product generating more than 1.89% of revenue, indicating healthy product diversification with no SKU concentration risk.

E. KPI Summary Table

KPI	Value	KPI	Value
Total Revenue	£8,911,407.90	Avg Order Value	£480.87
Total Quantity	5,167,812	Avg Unit Price	£3.12
Unique Customers	4,338	Avg Qty per Line	13.0
Unique Products	3,665	Unique Invoices	18,532

KPI	Value	KPI	Value
#1 Country	UK — £7,308,391 (82.01%)	#2 Country	Netherlands — £285,446 (3.20%)
#1 Product	Paper Craft Little Birdie — £168,469	#2 Product	Regency Cakestand — £142,593
Peak Month	November 2011 — £1,161,817	Trough Month	February 2011 — £447,137
Raw Records	541,909	Clean Records	397,884 (73.4%)
Cancelled Orders	9,288 removed	Null Customers	135,080 removed
Countries	38	Date Range	Dec 2010 – Dec 2011

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — 144,025 Rows of Dirty Data Mixed With Valid Sales

- Problem: 26.6% of raw records were invalid — 9,288 cancelled orders mixed with real sales, 135,080 transactions with no customer, 10,624 negative quantities from returns, and 2,517 zero-price test entries. Leaving any of these in would corrupt all revenue aggregations.
- Solution: Applied 4 sequential filter steps in Power Query in precise order — (1) Text Filter on InvoiceNo 'Does Not Begin With C', (2) Uncheck null in CustomerID, (3) Number Filter on Quantity Greater Than 0, (4) Number Filter on UnitPrice Greater Than 0. Each filter ran on the already-filtered dataset, maintaining data integrity throughout the pipeline.

Challenge 2 — No Revenue Column in Raw Dataset

- Problem: The dataset had Quantity and UnitPrice as separate columns but no Revenue field. Without a Revenue column it was impossible to create a SUM-based Total Revenue DAX measure — the core KPI of the entire dashboard.
- Solution: Used Power Query's Add Column → Custom Column to create Revenue = [Quantity] * [UnitPrice] after all cleaning steps, ensuring the multiplication only ran on valid positive rows. Then changed the column type to Decimal Number. The Revenue column appeared immediately as a summable field in the DAX measure editor.

Challenge 3 — StockCode Mixed Alignment Indicating Wrong Data Type

- Problem: StockCode column showed mixed alignment — some values left-aligned (text like '85123A') and some right-aligned (numeric like '71053') — indicating Power BI had assigned 'Any' type, causing inconsistent visual display and potential filter failures.
- Solution: Changed StockCode column type to Text in Power Query. This forced all values to left-align consistently regardless of whether they contained letters or not — preventing any grouping mismatch where '71053' (number) and '71053' (text) would be treated as different values.

Challenge 4 — All Revenue Visuals Showing Error After Loading

- Problem: After loading, the Total Revenue KPI card showed 'Error fetching data for this visual' and all charts using Total Revenue displayed black error boxes — while Quantity, Customer, and Product measures worked fine. This indicated a data type issue specific to the Revenue column.
- Solution: Returned to Power Query via Home → Transform Data → clicked the Revenue column header → changed type from 'Any' to 'Decimal Number' → clicked Replace current. The 'Any' type was set by the final 'Changed Type1' step which had overridden the Revenue column type. After Close & Apply, all visuals instantly populated with correct data.

Challenge 5 — KPI Cards Showing 9M Instead of 8,911,407

- Problem: Power BI's new Card visual defaulted to 'Auto' display units — abbreviating £8,911,407 as '9M' (rounding up), hiding the precise figure needed for a portfolio-quality dashboard that showcases analytical attention to detail.
- Solution: Clicked each KPI card → Format pane → Callout section → Value subsection → Display units → changed from Auto to None. Applied to all 4 cards individually to show exact figures: £8,911,407 | 5,167,812 | 4,338 | 3,665.

11 SKILLS LEARNED & TAKEAWAYS

- Multi-Step Data Quality ETL: Executed 4 sequential filter operations in Power Query — each building on the previous — removing 144,025 dirty rows while maintaining 397,884 valid records; learned the importance of filter ORDER when each step operates on the previously filtered result.
- Calculated Columns vs Measures: Created Revenue as a Power Query calculated column (row-level, stored) vs DAX measures (dynamic, filter-context-aware) — understanding when each is appropriate: columns for row-level math, measures for aggregated KPIs.
- DISTINCTCOUNT() for Cardinality: Used DISTINCTCOUNT() for Unique Customers and Unique Products — learning that DISTINCTCOUNT counts unique values while COUNT counts non-blanks and COUNTROWS counts rows — each producing different results on the same data.
- DIVIDE() for Safe Division: Used DIVIDE([Total Revenue], DISTINCTCOUNT([InvoiceNo])) instead of direct division — learning that DIVIDE() handles zero-denominator cases gracefully returning BLANK() instead of throwing an error.
- Data Type Impact on Visuals: Experienced how 'Any' data type on Revenue caused every visual using it to fail — while measures using other correctly-typed columns worked fine. Learned to always verify the data type icon on every column before Close & Apply.
- UK Market Concentration Analysis: Developed the analytical skill of identifying concentration risk — 82.01% revenue from one country is both a strength (deep market penetration) and a risk (single-market dependency). Learning to frame data patterns as business insights rather than just numbers.
- Seasonal Demand Pattern Recognition: Identified Q4 seasonality (Sep–Nov = 35.4% of annual revenue) as an actionable operational insight — not just a chart observation — and recognized the practical implications for inventory planning and staffing strategy.

12 CONCLUSION

- Processed 541,909 raw online retail records through a structured 8-step Power Query ETL pipeline — removing 9,288 cancelled orders, 135,080 null-customer rows, 10,624 negative quantities, and 2,517 zero-price entries — achieving a 73.4% clean data retention rate with 397,884 rows loaded at zero errors.
- Created Revenue calculated column (Quantity × UnitPrice) in Power Query, corrected 2 data type errors (StockCode to Text, InvoiceDate to Date, Revenue to Decimal), engineered 5 DAX

measures, and built a 12-visual dashboard with country bar chart, monthly trend line, donut chart, world map, product table, and 2 cross-filtering slicers.

- Surfaced £8,911,407 total revenue, 82.01% UK market concentration (£7,308,391), November 2011 as peak month (£1,161,817 — Christmas surge), Paper Craft Little Birdie as #1 product (£168,469), and Avg Order Value of £480.87 — actionable insights identifying both UK dependency risk and Q4 seasonal demand patterns.
- Identified Q4 2011 (Sep–Nov) generating £3,153,973 (35.4% of annual revenue in 3 months), Netherlands and Germany as highest-potential international expansion markets, and White Hanging Heart T-Light Holder as most frequently ordered product (1,971 invoices) — all extracted from a single dashboard without writing any code.
- Delivered a custom Digital Commerce JSON theme (teal/navy canvas, white card borders, orange data accents) — demonstrating complete Power BI workflow from raw Excel ingestion through multi-step ETL, DAX modelling, visual design, and portfolio-quality branded presentation. Project 3 of 9.